DunesForge - notes for beta testers

Version 0.85 beta. Thank you for testing DunesForge. Most things work, some will not, and your reports decide what is fixed before version 1.0.

Questions: the manuals first, then Discord

The manuals that come with the add-on should explain everything you need. When something is unclear or does not work, please look there first:

- Getting Started - from installation to your first dune field in ten minutes.
- Troubleshooting - the most common problems and what to check first.
- QuickStart - the full guide with tutorials.
- Help for one control: right-click it and choose **Explain this DunesForge option**. The **?** icon on each panel opens its chapter.

The manuals are PDF files in the add-on's docs folder; the buttons at the top of the DunesForge panel open them. If the answer is not there, ask me on Discord. Please also tell me what you were looking for, so it can be added to the manuals.

Installing and updating

You need **Blender 5.1 or newer**. Older versions, from 4.2 on, may work, but they have not been thoroughly tested.

1. **Edit > Preferences > Get Extensions**, the small arrow in the top-right corner, **Install from Disk**. Pick the DunesForge ZIP. Do not unpack it.

2. **Restart Blender.** This is also needed after every update.
3. Press **N** in the 3D Viewport and open the **DunesForge** tab. The version is shown in the panel, for example **DunesForge build 0.85.5 beta**. If the panel says **RESTART BLENDER**, the old version is still running.

Step by step: Getting Started.

What to try

- Open two or three demo scenes and press **Play / Pause**.
- Make a dune field on your own terrain.
- With an NVIDIA card: click **Install CuPy GPU Backend** (it downloads about 1 to 2 GB), restart Blender and check that the panel says **Running on GPU**.
- Water, snow, painting, rendering: whatever you would use in a real project.
- Tell us also when something was confusing or did not do what you expected. That is a bug too.

Slopes too steep or piling up

Where sand keeps arriving -- a **Sand Source**, sand blown against a rock or a wall -- the slopes can grow much steeper than real sand stands, up into tall spikes and towers. Since 0.85.15 a **Sand Source** switches a light settle on by itself (**Auto Settle Fed Piles**, every 4 steps). If slopes still pile up, set **Settle Slopes** in the **Physics** panel (switch **Interface** at the top of the panel to **Advanced** to see it):

- **Settle Slopes Every N**: **1** (a settle burst after every solver step).
- **Settle Rounds**: **10**. Raise it towards **20** only if the slopes are still too steep.

Measured on a pile fed by a Sand Source: without Settle the steepest slope reached 60 to 78 degrees; with these settings it stays at about 34 degrees, the natural angle of sand. On the GPU it

costs about 25 to 35 % more time per step with 10 rounds, and about 50 to 65 % with 20.

This is the cheaper option. **Avalanche Trigger** (also in **Physics**) solves the problem the natural way: an over-steep face lets go as a real slide, so the slope cannot pile up far above the natural angle. It costs more (in our measurements up to twice as much as Settle), so keep it for shots where you want to see the slides.

Wind solver modes

Wind Model in the **Wind** panel decides how the wind is calculated:

- **Analytic (Jackson-Hunt)** - the default. Fast, and it grows correct dunes, but the wind does not really flow: it cannot go round rocks, swirl behind them or speed up through a gap.
- **Flow 2D (solved)** - solves the air in plan view, so the wind goes round rocks and walls, speeds up in canyons and leaves a turning wake behind obstacles. It costs extra time; **Flow Quality** and **Solve Every** set how much. By default only obstacle objects block the wind. For rock walls, canyons or mesas in the heightmap, set **Wall Height** to roughly their height (the flow demos use 2 m) and the terrain deflects it too. Keep it above the height of your dunes: wind goes over a dune, not round it, and at 0.5 m one cell in nine of a dune field turned into a wall and the crests broke up into steps.
- **Flow 3D** - the **3D Wind (EXPERIMENTAL)** checkbox inside the Flow 2D settings (Interface: **Advanced**). The wind also climbs over terrain and objects, so the wind shadow behind a dune, a rock or a building gets its real length -- 5 to 11 times its height instead of a fixed 4. It is slow, still being finished, and needs an NVIDIA GPU; on the CPU it is only approximate.

For the most realistic behaviour use **Flow 2D**. For even more accuracy turn on **Flow 3D**, and expect it to take much longer.

Shallow water is a preview

Shallow water -- the flowing-water simulation and the demos marked **(Experimental)** -- is a preview. It needs deeper integration with the rest of the simulator and will probably not be part of the first release (1.0); it is planned for a version after it. The simple water level, rain and moisture are not affected. Try it and tell us what you think, but expect it to change.

Known limitations of this beta

- The supported minimum is Blender 5.1. Tested on Windows with Blender 4.4 to 5.3; versions before 5.1 may work, but they have not been thoroughly tested. Linux and macOS have not been tested yet, so reports from them are especially welcome.
- GPU acceleration needs an NVIDIA card. AMD, Intel and Mac computers run on the CPU, which is slower.
- Flowing water is part of the running simulation and is not saved in the .blend file; the file keeps only the water the demo started with. After opening the file, click **Initialize Simulation** to bring that water back.
- Some demos take a few minutes to build.
- Add-ons that bring their own CUDA libraries, for example BlendLuxCore, can stop the GPU backend from working. See Troubleshooting.

Reporting a problem

1. Click **Report a Bug** at the top of the DunesForge panel. It opens the bug form on GitHub. You can also write to **forgeplugins@gmail.com** or ask on Discord.
2. **Attach the log file.** Click **Open Log Folder** in the main panel and attach **tydunes.log**.
3. Give the version from the panel, your Blender version and your graphics card.

4. Describe what you did, what you expected and what happened. A screenshot or a small .blend file that shows the problem helps a lot.

What the log contains

The DunesForge and Blender versions, the graphics card, the DunesForge buttons you pressed, the settings you changed before pressing them, error messages and the name of each .blend file you opened. It does not contain your scenes or any other files. Error messages can include folder paths, which may show your Windows user name. You can open tydunes.log in any text editor and read it before sending.